

Intro to Psychology

book notes!

psychology: scientific study of mind, brain and behavior, because common sense fails

mind = mental activity

brain = cause of biochemical processes that result in mental activity

behavior = totality of observable human or animal actions

ch 1.2

critical thinking is important when looking at research, be aware of your own biases!

amiable skepticism

ch 1.3

common biases:

- confirmation bias/ignoring evidence: people favor information that supports their beliefs
- seeing casual relationships that do not exist: Correlation $\neq$ causation!
- hindsight bias: accepting after-the-fact explanations, it's easy to explain things that happened in the past which gives us the false belief that we can predict the future.
- taking mental shortcuts: people use heuristics (simple rules) to make decisions because it doesn't take too much effort, but it can lead to inaccurate judgement and biased outcome.
- Dunning-Kruger effect: people lack the ability to evaluate their own performance in areas where they have little expertise
most people are unaware of their weaknesses

ch 3.15

environmental factors can affect gene expression

genome = 'blueprint'

genes are segments of DNA-strands $\rightarrow$ make polypeptides $\rightarrow$ proteins

the human genome project is trying to map ~~at~~ the sequence of human DNA

the number of genes are of less importance than the way those genes are expressed and regulated.

there are dominant genes and recessive genes

two influences on phenotype:

- genetics
- environment (nurture)

polygenic traits are influenced by many genes

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behavioral genetics studies how genes and environment interact to influence

psychological activity.

genes lay the groundwork for many human traits, in combination with the environment

→ twin studies and adoption studies are used to research which traits are inherited.

monozygotic = identical, dizygotic = fraternal

heredity: transmission of characteristics

heritability: statistical estimate of the extent to which variation in a trait within a population is due to genetics.

epigenetics: gene expression can be altered by environment and this can be passed on to future generations

optogenetics: research method to control when a neuron fires

cells → nucleus → 46 chromosomes → DNA → double helix with complementary chains

conditions of natural selection:

- variation among individuals of a population
- individuals with a certain trait survive + reproduce better
- the trait is passed on to offspring

evidence for evolution: fossil record, resemblance between genomes and pseudogenes and distribution of species

continuous interaction between genes and environment

ch 3

central nervous system - brain and spinal cord

peripheral nervous system - other nerve cells

↳ somatic and autonomic (animal en autonoom)

neurons form neural networks to communicate

sensory neurons detect information

motor neurons direct muscles

interneurons facilitate communication between the two

reception → integration → transmission

dendrites → cell body → axon → terminal buttons

chemical communication happens in the synapse

resting membrane potential: inside of the neuron is more negative. otherwise polarized

excitatory signals depolarize the cell → increased firing

inhibitory cells hyperpolarize the cell $\rightarrow$ decreased firing
myelin sheath is made up of glial cells, with nodes of Ranvier
in between. it increases the speed of the action potential
depolarization $\rightarrow$ repolarization $\rightarrow$ relative refractory period
during the absolute refractory period, the ion channels can't open again

neurotransmitters are chemicals in the terminal buttons

they bind to receptors

neurotransmitter's effect is terminated by:

- reuptake: neurotransmitter goes back into terminal buttons
- enzyme deactivation: destroyed
- autoreception: binding with receptors

agonist enhance actions of neurotransmitters, antagonists inhibit them

phrenology studies bumps in the skull, 19th century science by Gall

Broca's area: part of the brain on the front left, essential to speaking

brain lesion methodology: using brain injury to discover functions of the brain

other brain research methods:

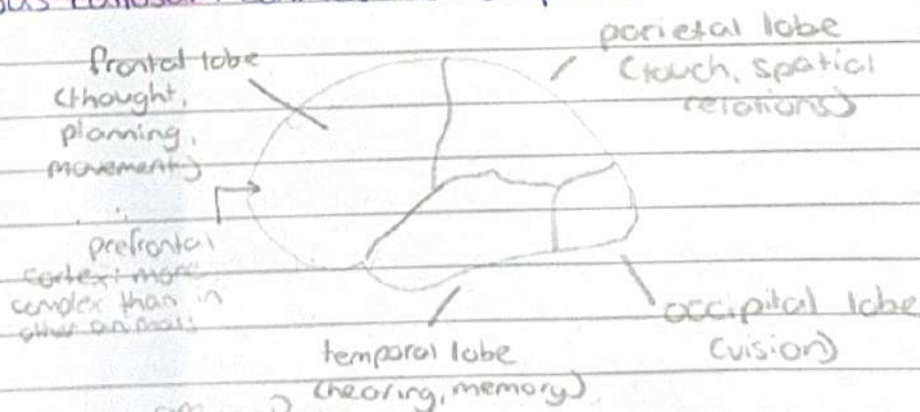
- EEG: electrodes on scalp pick up electrical activity
 $\rightarrow$ reflect all brain activity, can't isolate specific response

- ERP: ~~many~~ average of many trials

functional brain imaging to monitor activity in areas of the brain

- PET: by injecting a radioactive substance in the bloodstream (tracer)
- fMRI: uses brain's blood flow to map the working brain (with changes in oxygen level)
 $\rightarrow$ downside: not sure which regions are 'necessary'
- TMS: disrupts neural activity in a specific brain region

forebrain is made up of cerebral cortex and underlying subcortical areas
corpus callosum connects the hemispheres



Freud Taking Off his Pants

Split brain: when corpus callosum is cut so the hemispheres can't interact. Results in two half brains

left side (language) can verbalize, right side can perceive
left side is the interpreter

left-brainers and right-brainers are a myth!

each hemisphere is capable of carrying out most cognitive processes, though most of the time both are used.

insula and subcortical regions lie under the cerebral cortex

↳ insula: inside the lateral fissure, for taste, pain, perception, empathy

↳ thalamus: receives information and relays to the cortex

↳ hypothalamus: regulation of bodily functions

↳ hippocampus: formation of memories

↳ amygdala: process emotional information

↳ basal ganglia: planning and production of movement

spinal cord becomes brain stem for basic functions of survival

cerebellum connected to brain stem, for motor function

Sympathetic nervous system prepares for action

parasympathetic nervous system returns body to a resting state

endocrine system uses hormones to communicate, a lot slower
via systems hormones get released through signals to the pituitary gland

plasticity: the brain's ability to change in response to injury or experience
decreases with age

'cortical reorganization'

ch 4
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consciousness: one's moment-to-moment subjective experience of the world
experiences of sensation are subjective: qualia

consciousness is like a continuous stream

consciousness is limited because you cannot focus on a lot of things at the same time

attention is selective

endogenous attention: intentionally directing your attention

exogenous attention: attention driven by stimulus or event

priming: response to a stimulus is influenced/facilitated by recent experience (it is unconscious)

subliminal perception is too short to reach the consciousness, but can still subtly influence behavior, especially when it invokes emotion or motivation
automatic processing happens without much attention
controlled processing is slower and requires attention

3 ways of naturally altering waking consciousness:

- meditation (concentrative or mindfulness)
- runner's high or religious ecstasy cause shifts in consciousness
- flow is an activity that is so enjoyable, you do it even though it has no other consequences. (immersion in an action)
- hypnosis: social interaction in which a person experiences changes

sleeplessness is the gene that influences sleep

Suprachiasmatic nucleus is a region in ~~the~~ the hypothalamus which detects light, then sends signals to the pineal gland which secretes melatonin?
EEG show beta waves when people are awake, and alpha waves when they focus their attention or close their eyes. In sleeping, there's theta waves. In deeper sleep, there's bursts of ~~energy~~ activity called sleep spindles or large waves called k-complexes. Deep sleep is slow-wave sleep and has delta waves

REM sleep = paradoxical sleep because sleeping body with active brain
there's different regions of the brain active during REM- and non-REM sleep
manifest content disguises latent content

activation-synthesis hypothesis - dreams are like side effects of mental processes produced by random neural firing

3 theories why we sleep:

- restorative theory, growth hormone is released more + by-products are cleaned up (to rest and repair)
- circadian rhythm theory, to be safe from danger
- facilitation of learning theory, neural connections get strengthened

insomnia - sleep disorder

obstructive sleep apnea - disorder in which people stop breathing during sleep.

narcolepsy - disorder in which people experience excessive sleepiness during waking hours.

traumatic brain injury - impairments in mental functioning after blow or very sharp movement of the head.

in a coma, brain imaging can be used to identify consciousness
unresponsive wakefulness syndrome still shows brain activity, brain death doesn't.

types of drugs: stimulants (increase behavioral + mental activity),
depressants (decrease behavioral + mental activity), opioids reduce pain,
hallucinogens alter thoughts/perception

ch 5

Sensation: detection of stimuli and transmission to the brain

perception: further processing, organization, and interpretation of sensory information

bottom-up processing: each sensory aspect of a stimulus is processed and then built up to perception

top-down processing: knowledge, expectations or past experiences shape the interpretation of sensory information (priming/context)

transduction is the translation of stimuli, using sensory receptors, brain needs qualitative and quantitative information] into neural signals

absolute threshold: minimum intensity required to experience a sensation

difference threshold: smallest difference between two stimuli that you notice

Weber's law: difference threshold depends on the proportion

signal detection theory: detecting is a subjective process, requires judgement

response bias is how the participant reports in an ambiguous trial

sensory adaptation - decrease in sensitivity to a constant level of stimulus

synesthesia is a cross-sensory experience

light → cornea → pupil/iris → lens → retina → optic nerve → brain's visual cortex

rods are used for peripheral and night vision

cones are for seeing colour and detail

most cones in the fovea (focus)

! ganglion cells are the first neurons in the visual pathway, then optic nerve

2 processing streams: ventral stream from occipital lobe to temporal lobe for 'what' stream (recognition and perception)

and dorsal stream from occipital lobe to parietal lobe for 'where' stream (spatial perception)

trichromatic theory: 3 types of cones cause us to see colour

opponent-process theory: there are opponent colours, receptors for one colour can become fatigued if you look long enough, which creates an afterimage in a different colour

colour categories: hue, saturation and lightness

experience can inform how we identify figures and distinguish them from the background.

Gestalt theory: brain uses principles to group sensory information

Gestalt grouping laws: proximity, similarity, good continuation, closure, illusory contours, common fate

object constancy: perceiving objects correctly even when sensory data is misleading

fusiform gyrus is critical for perceiving faces

depth cues: - binocular, from both eyes, give internal cues

- monocular, from each eye, e.g.

binocular disparity calculates the difference between the eyes' views

convergence: the convergence of the eye muscles can also be used to perceive distance.

monocular depth cues: occlusion, relative size, familiar size, linear perspective, texture gradient, position relative to horizon, and motion parallax

ch6

learning: relatively enduring change in behavior resulting from experience

- nonassociative learning: occurs after repeating single stimulus

- associative learning: through conditioning, linking two events that happen after each other.

- social learning: acquiring behaviour and predictive associations through interaction with others

nonassociative

↳ habituation: decrease in behavioural response after repeated exposure to stimulus (can still perceive but become unaware)

↳ dishabituation: increase in response because of a momentary change in something familiar (become aware again)

↳ sensitization: increased response to a stimulus after repeated exposure, leading to heightened responsiveness

↳ desensitization: extinction of sensitization

classical conditioning (Pavlov) - you learn that one stimulus predicts another

starts CS
neutral stimulus

unconditioned response (UR) - occurs without training, salivating when seeing food.

unconditioned stimulus (US) leads to UR

conditioned stimulus ^(CS) is ~~trained~~ actually unrelated, but used for training

conditioned response (CR) is the learned response

the stimuli need to occur together in time

acquisition: formation of an association between a conditioned stimulus and an unconditioned response stimulus

extinction: the CR slowly disappears if the CS no longer predicts the US through spontaneous recovery, it can make a comeback, but temporary

Pavlov's - Wagner model: learning is determined by the extent to which an unconditioned stimulus is unexpected or surprising

positive prediction error - stronger stimulus, strengthens CS-US association

negative prediction error weakens CS-US association

difference between expected outcome and actual one

stimulus generalization: similar stimuli produce the CR

stimulus discrimination: differentiate between similar stimuli

- second-order conditioning is when a CS gains its own value and can get its own CR

operant conditioning: learning actions from consequences

Thorndike's law of effect: behaviour that leads to satisfying state is likely to occur again.

behaviorism: environmental influences on observable behaviors

reinforcer: stimulus that increases the likelihood of repetition

positive reinforcement: reward, something is added

negative reinforcement: removal of unpleasant.

positive punishment gives an unpleasant stimulus to decrease behaviour

negative punishment removes a pleasant stimulus to decrease behaviour.

rewarding good behaviour > punishing bad behaviour

in shaping, you reinforce behaviour that's similar to the behaviour you want until it's accomplished.

primary reinforcers are necessary for survival

secondary reinforcers do not satisfy biological needs

temporal discounting: value of a reward diminishes over time

continuous reinforcement is for fast learning, not common

Partial reinforcement is intermittent

- fixed interval: reinforcement after certain amount of time
- variable interval: reinforcement after irregular amount of time
- fixed ratio: reinforcement after number of responses
- variable ratio: reinforcement after ~~un~~ unpredictable number of responses

partial-reinforcement extinction effect: behaviour is more likely to persist through partial reinforcement than through continuous reinforcement

equipotentiality suggests that any conditioned stimulus paired with any unconditioned stimulus should result in learning (not really true)

animals learn something faster if it's similar to their evolutionary history.

biology constrains learning actions; conditioning is most effective when the association is similar to the animal's built-in predispositions

biological preparedness makes it more likely to fear potentially dangerous stuff.

more dopamine → better reward

phobia: acquired fear that is out of proportion to the real threat

fear conditioning: learning to be scared of neutral objects

modeling: imitation of observed behaviour

vicarious learning: learning consequences by watching others

instructed learning (in humans): verbally learning associations and behaviour

→ observational learning

ch 7 memory: ability to store and retrieve information

amnesia: inability to remember

retrograde amnesia: inability to remember past events

anterograde amnesia: inability to form new memories

different types of memory rely on different brain regions

implicit memory: 'unconscious' memory, procedural memory (skills and habits) and classical conditioning

explicit memory: 'declarative' memory, episodic memory (past experience at a certain time and place) and semantic memory (facts and knowledge)

3 stages of processing: encoding, storage, retrieval

encoding: perception of a stimulus gets ~~encoded~~ transformed into a memory
more easily encoded if there's a visual and if it's thought about more deeply or by chunking (breaking down information into meaningful units)

mnemonics are strategies to improve memory by organizing

Storage: 3 types of memories

Sensory memories are fleeting; iconic memory is visual, echoic memory auditory

Short-term memory = working memory, what we are consciously focusing on. rehearsal keeps it in your mind. capacity ≈ 7

long-term memory has unlimited capacity

Serial positioning effect: things at the beginning and end of a list get remembered better.

consolidation: forming lasting connections that represent long-term memory
'cells that fire together wire together'

long-term potentiation: strengthening synaptic connection

slow consolidation is good because only important memories are stored.

flashbulb memories: vivid, detailed memories for shocking and consequential events. the memory for this is inconsistent, but most people are confident.

Sometimes memories are reconsolidated when retrieving again.

2 functions: memory updating and memory strengthening

a retrieval cue helps someone recall a memory

encoding specificity principle - any stimulus encoded along with an experience can later trigger a memory of the experience

context-dependent memory - based on external stimuli

state-dependent memory - based on internal stimuli

prospective memory - remembering to do something at some future time

retrieval-induced forgetting - retrieving an item from long-term memory

impairs the ability to recall a related item in the future

savings: difference between original learning and relearning

proactive interference: old information inhibits the ability to remember new information

retroactive interference: new information inhibits the ability to remember old information.

blocking: temporarily unable to remember something

absentmindedness: the inattentive or shallow encoding of events

persistence: unwanted memories are remembered

memory bias: changing of memories over time so that they become consistent with current beliefs, knowledge or attitude

Source misattribution: time, place, person or circumstance is misremembered

Source amnesia: forgetting where information was encountered

Cryptomnesia: people think they came up with something

Suggestibility: people tendency to develop biased memories when provided with misleading information

ch 8

Cognition: mental processes involved in acquiring knowledge and comprehension through thought and experience

thinking: mental manipulation of representations of knowledge about the world.

2 basic types of mental representation:

- analogical representations have some characteristics of what they represent, usually images
 - symbolic representations are abstract, usually words, numbers or ideas
- categorization is grouping things together

concept: category/class of related items

2 leading models how people form concepts:

- prototype model: think of 'best example' in a category
- exemplar model: all examples form the concept

knowledge of how to behave in different settings relies on schemas

script: schema that directs behavior over time within a situation

Stereotypes: cognitive schemas that allow for easy, fast processing of information about people based on their membership in certain groups. they persist because they minimize required attention

decision making: selecting among alternatives

problem solving: overcome obstacles to reach a goal state

heuristics - mental shortcuts in thinking

anchoring: when people focus on the first thing they can think of during decision making

framing is the way information is presented

loss aversion - people are more concerned with loss than benefit

availability heuristic: tendency to make a decision based on the first thing that comes to mind.

representativeness heuristic tendency to place a person/object in a category if they're similar to the prototype

affective forecasting: predicting how you'll feel about things in the future

endowment effect: tendency to value things we own more than we'd pay

ways of problem solving:

- organizing into subgoals
- revise mental representation: restructuring the problem
- often use mental sets (previous strategies) in problem solving
- functional fixedness: fixed ideas about an object's function
- algorithm: guideline that will always yield the correct answer
- sudden insight

words consist of morphemes: smallest units that have meaning
that consist of phonemes: sounds

people speak by forcing air through the vocal chords

aphasia: deficit in language comprehension / production because of brain injury

expressive aphasia / Broca's aphasia: interrupts ability to speak

receptive aphasia / Wernicke's aphasia: inability to understand meaning

linguistic relativity theory: language determines thought (not true)

children often use telegraphic speech

parents do not use operant reinforcement to teach language

surface structure: sound and order of words

deep structure: implicit meanings of sentences

Chomsky → humans are born with a language acquisition device

humans must possess a biologically endowed sensitivity to the perception and organization of language, because sign language and spoken language are acquired in very similar ways.

creole: language that evolves from mixing languages

social communication, exposure and communication during the critical periods of development are ~~peress~~ essential to language acquisition

2 schools of thought on how to teach reading

phonics: association between letters and sounds

whole language approach: learning the meaning of words

intelligence: ability to use knowledge to reason, make decisions, etc.

psychometric approach uses tests

intelligence as collection of high-level mental processes?

mental age: child's intellectual standing compared to the average

intelligence quotient: $\text{mental age} / \text{chronological age}$

are they reliable (stable over time) and valid (measure the right things?)

general intelligence: the idea that one general factor underlies intelligence

2 types of intelligence:

- **fluid intelligence**: ability to think logically without prior knowledge

- **crystallized intelligence**: knowledge acquired through experience

other theory: each person has a unique pattern of intelligences

triarchic theory: there's analytical, creative and practical intelligence

emotional intelligence: consists of managing emotion, using emotions to guide thoughts and actions, recognizing other people's emotion, and understanding emotional language

nature and nurture are intertwined in the development of intelligence

environmental influences affect human intelligence

enriched environment is better

new book

ch 1

empiricist: base on systematic ~~behavior~~ observations

evidence based treatment is supported by empirical research

theory: (set of) statements that describes general principles about how variables relate to one another

hypothesis: prediction

data: set of observations

preregistered hypotheses are made before the data collection

empiricists are not justified in making generalizations about phenomena they have not observed

falsifiability: a good theory has hypotheses which could prove it to be incorrect.

4 norms for scientists:

- **universalism**: independent of the researcher's credentials or reputation

- **communality**: created by and belongs to a community

- **disinterestedness**: strive to discover truth

- **organized skepticism**: question everything

applied research with practical problem in mind

basic research for general knowledge

translational research is the bridge

- did not do ch. 2 -

Ch 3

Variable: attributes that varies and has at least 2 levels

a **constant** has only one level in the study

measured variable: the levels are observed and recorded

manipulated variable is controlled by the researcher

constructs / conceptual variable: on abstract level, name of concept

operational variables: turn it into a measured / manipulated variable

conceptual definition: theoretical definition of the construct

3 types of claims (arguments)

- **frequency claims:** describe a rate or degree of a variable

- **association claims:** argues that one level of a variable is likely to be associated with a particular level of another variable

↳ **correlation studies** test the relationship between two variables

positive, negative or zero

- **causal claim** argues that one of the variables is responsible for changing the other

validity: the appropriateness of a conclusion or a decision

to evaluate how well a study supports a **frequency claim:**

→ **construct validity:** how well a conceptual variable is operationalized

↳ measured reliably and different levels correspond accurately to true differences

→ **external validity:** concerns generalizability, how well do the participants represent the intended population

→ **statistical validity:** are the statistical conclusions precise, reasonable and replicable?

confidence interval / margin of error of the estimate

For association claims:

→ **construct validity:** how well both variables are measured

→ **external validity:** can it generalize to other populations / places / contexts
other:

→ **internal validity:** the extent to which one variable is responsible for changes in another, rather than some other variable

criteria for causation:

- **covariance:** extent to which two variables are observed to go together

- **temporal precedence:** the cause has to come before the effect

- **internal validity:** alternative explanations should be eliminated

to support a causal claim, an experiment should be conducted →

manipulating the 'cause' variable and measuring the 'outcome' variable

↑
independent variable

↑
dependent variable

random assignment increases internal validity by using a random method to assign participants into different experimental groups. researchers usually have to prioritize one of the validities, it shouldn't be faulty just because it didn't meet one of the validities.

ch 11

a one-group, pretest/posttest design doesn't have a control group
potential threats to internal validity: in one-group pretest/posttest design

- maturation threat: change in behaviour that emerges more or less spontaneously over time.
- history threats: another factor that affects most members of the treatment group. Systematically (in the same direction)
- regression threat: if a group's average is unusually high, it will likely decrease
- attrition threats: reduction of participant numbers. becomes a problem if only a certain type of participant drops out
- testing threat: change in participants because of taking multiple tests.
- instrumentation threat: when a measuring instrument changes over time

combined threats:

- selection-history threat: other factor affects only those at one level of the independent variable.
 - selection-attrition threat: only one group experiences attrition
- potential internal validity threats in any study
- observer bias: when researcher's expectations influence their interpretation of the result. (also construct validity)
 - demand characteristics: participants who know what the study is about change their behaviour
 - ↳ to avoid: double-blind study, participant and researcher don't know who's the treatment or control group
 - masked design: only researchers don't know the groups
 - placebo effect: people improve because they believe they get a valid treatment.
 - ↳ double-blind placebo control study: neither researcher nor participant know who's taking real treatment and who's taking the placebo.

ch 5 conceptual definition: the researchers definition of the variable in question at theoretical level

common types of measures:

- self-report measures: answers about yourself
- observational measures: recording observable behaviours
- physiological measures: recording biological data

all variables must have at least 2 levels:

- categorical variables / nominal variables: categories
- quantitative variable: numbers that represent different levels
 - ↳ ordinal scale: ranked order, but intervals can be unequal
 - ↳ interval scale: equal intervals, without a true zero
 - ↳ ratio scale: equal intervals, with a true zero

construct validity of a measure has two aspects:

- reliability: how consistent the results are
 - ↳ test-retest reliability: same score every time the test is done
 - ↳ inter-rater reliability: same score no matter the measurer
 - ↳ internal reliability: consistent answers no matter the phrasing of the question → Cronbach's alpha

correlation coefficient r indicates how close dots on a scatterplot are to a line drawn through them

strength of the relationship is shown by how scattered the dots are

ch 6 survey & poll: method of posing questions to people online

possible formats: open-ended questions, forced-choice questions, likert scale, semantic differential format

leading questions may influence the way people answer

double-barreled questions ask two questions in one

negatively worded questions can also reduce construct validity

question order could also influence answers

people could use response sets when answering ~~question~~ surveys:

- ↳ acquiescence (yea-saying)
- ↳ fence sitting (neutral answering)
- ↳ faking good or faking bad

people may not be able to explain why they acted as they did accurately

observational research: researchers watching people or animals and systematically recording how they behave or what they are doing.
construct validity of observations can be threatened by:

- observer bias: observer's expectation influences their interpretation
- observer effect: observer changes in behaviour of participants in the direction of the observers expectation
'codebooks' to fix this, or masked design, so observers are unaware of the purpose of the study
- reactivity: change in behaviour when participants know someone is watching
solutions: blend in, wait it out, measure the behaviour's results

ch 7

census: observations containing all members of a population

biased sample: some members of the population of interest have much higher probability than others.

unbiased samples: all members have an equal chance of being included

convenience sampling: using people who are easy to contact and readily available

self-selection: only volunteers participate

ch 8

bivariate correlation/association: association that involves two variables

there are categorical and quantitative variables

bar graph $\leftarrow$ $\uparrow$ represent through scatterplot

$\hookrightarrow$ to show the mean

questions for association claims:

- construct validity: how well was each variable measured?
- statistical validity: how well do the data support the conclusion
 $\hookrightarrow$ effect size: strength of a relationship between two or more variables
how precise is the estimate?

a statistically significant correlation (95% CI doesn't include zero) is unlikely to have come from a population in which the association is zero.
has it been replicated?

could outliers be affecting the association?

is there restriction of range?

makes correlation appear smaller if there's not a full range of scores

is the association curvilinear?

curvilinear is a relationship in a curve

- **internal validity**: can we make a causal inference from an association?
careful of the directionality problem: don't know which variable came first, so we can't say causation. → = temporal precedence
third-variable problem as well: there might be an alternative explanation → = internal validity

spurious association: a bivariate association that is attributable only to systematic mean differences on subgroups within the samples; the original association is not present within the subgroups (length height and hair length)

- **external validity**: to whom can the association be generalized?

moderator: a variable that, depending on its level, changes the relationship between two other variables

Lack of external validation is not that big of a problem, if it fulfills the other three, the generalizability can be left for future investigation

ch 9 **multivariate designs** involve more than two variables

advanced correlation techniques to get closer to making a causal claim:

longitudinal designs, multiple-regression analyses, and 'pattern and parsimony'

longitudinal design: measuring same variables in same people at several times

cross-sectional correlation: to see if two variables at the same point of time are correlated

autocorrelations determine the correlation of one variable on two occasions

cross-lag correlations is what researchers are most interested in.

shows whether the earlier measure of one variable is associated

with the later measure of the other variable, to establish temporal precedence

→ provide evidence for causal relationship by:

1. **covariance**: when variables are correlated and the 95% CI doesn't include 0

2. **temporal precedence**: because the variables are measured at different points in time

3. **internal validity**: third variable can't be ruled out

it's a good option when an experiment is not practical or ethical

ch 10 in an **experiment**, a researcher **manipulates** at least one variable and **measures** another. a **manipulated variable** is controlled. **measured variables** are records of behaviour or attitude

independent variable = manipulated

the conditions are the 'levels'

dependent / outcome variable = measured

control variables are held constant during the experiment

an experiment is better than personal experience because it has a control comparison group/condition

control group: level of independent variable that's the neutral condition

treatment group is the other level

placebo group when exposed to an inert treatment

experiments have temporal precedence, correlation studies don't (snapshot)

confounds are possible alternative explanations

to internal validity:

- design confound: a second variable varies systematically along with the independent variable (so alternative explanation)

systematic variability: when the levels of a variable coincide in some predictable way with experimental group membership (potential confound)

unsystematic variability: levels fluctuate independently of experimental group membership. (no design confound)

- selection effect: when the kinds of participants in one level of the independent variable are systematically different from those in the other

can be avoided with random assignment

matched groups can also work to make it equal

- order effects (see within-groups)

independent-groups design: separate groups of participants in different levels
= between-subjects / groups

within-groups design: each participant is presented with all levels of the independent variable.

↳ 2 basics:

- posttest-only design (= equivalent groups): participants are randomly assigned to independent variable groups and are tested once on the dependent variable. (satisfy all 3 criteria for causation)

- pretest/posttest design (= equivalent groups): participants are randomly assigned and are tested on the dependent variable before and after exposure to the independent variable. can be used to make sure groups are equal, and to track progress

↳ also 2 basics:

- repeated-measure design: participants are measured more than once.

after exposure to each level of the independent variable

- concurrent-measure design: participants are exposed to all levels at (roughly) the same time, dependent variable is one attitudinal / behavioral preference

within-group ensures the participants are equivalent (bc they're the same), and researchers can make more precise estimates bc less unsystematic variability, and less participants needed.

it does threaten internal validity because of order effect: exposure to one condition changes participant responses to a later condition.

like practice effects (fatigue effects), if a participant gets better at a task / bored / tired, or carryover effect, when some sort of contamination carries over from one condition to the next, to fix this, researchers counterbalance, so present the levels of the independent variable in a different sequence

↳ full counterbalancing: all possible conditions are represented, usually only used when there aren't many levels

↳ partial counterbalancing, if there are more levels, so only some of the possible condition orders are represented

↳ random order

↳ Latin square, so each condition appears in each square at least once

↳ balanced Latin square

disadvantages of within-group: order effects, permanent change when teaching subjects something, or participants changing behaviour = demand characteristic.

to check how well independent variables were manipulated, a manipulation check (extra dependent variable) can be used

a pilot study uses a separate group of participants to do a simple study to confirm the effectiveness of the manipulations.

ch 11 if the 95% CI includes zero, it can be called a null result / effect

it can mean that the independent variable just doesn't affect the dependent one, or that the study just wasn't able to detect it reasons could be not enough between-group difference or too much within-group variability (weak manipulations / insensitive measures / ceiling effect (all scores are high) / floor effect (all scores are low))

another reason is noise: too much unsystematic variability within each group

measurement error can cause high within-group variability, or individual differences

Situation noise is external distraction

power in a study is the likelihood that it will return an accurate result, when the independent variable really has an effect, like adding more participants

ch 12

interaction effect / interaction: whether the effect of the original independent variable depends on the level of another independent variable
= a difference in differences

spreading interaction: 'especially', lines don't cross



factorial design: has 2 or more independent variables (factors)
most common factorial design is crossing, in which each possible combination of the independent variable is studied. this creates 'cells' of the combinations.

participant variables are selected, not manipulated, but they're often still called independent variables.

can be used to test external validity (does it generalize?).

there are 3 results to inspect in a factorial design

main effect (x2): the effect of one independent variable on the dependent variable, averaging over the levels of the other independent variable

marginal means is like the average over levels of the other independent variable

= 'overall effect'

so don't mention them

interaction effect: difference in differences, more important than main effect
can see in a graph if not parallel or in bar graph if you can draw non-parallel lines

factorial designs are used to test generalizability and to test theories
the experiment can be independent-group, within-group or mixed (one variable as independent group and one as within-group)

three-way interaction: the differences between differences is different

- testing more variables means you can study more interactions, which is more realistic.

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in quasi-experiments, researchers don't have full control and usually can't assign participants.

the independent variable is called the quasi-independent variable
4 common designs:

- nonequivalent control group posttest-only design: participants aren't randomly assigned, and test was done once after exposure
- nonequivalent control group pretest/posttest design: participants aren't randomly assigned and were tested before and after intervention
- interrupted time-series design: a variable is measured repeatedly before, during and after the 'interruption'
- nonequivalent control group interrupted time-series design: participants aren't randomly assigned, variable is measured repeatedly before, during and after the interruption.

internal validity is the main concern in quasi-experiments

↳ selection effect: are the participants systematically different?
can be controlled for with wait-list design, participants are randomly assigned times to get treatment

↳ design confound: outside variable can have an influence

↳ maturation threat: change could have emerged spontaneously

↳ history threat: an external event can influence the results

↳ regression to the mean

↳ attrition threat: systematic kinds of people drop out of a study

↳ testing / instrumentation threats

↳ observer bias, demand characteristics, placebo effects

reasons to use a quasi-experiment:

- real-world opportunities
- external validity
- ethical concerns

small-N design focus on getting a bit of information from a few cases
participants are treated separately and data for each individual are presented

they can still advance our knowledge because of experimental control,
and it gives the opportunity to study special cases

disadvantages internal validity, because multiple changes could result in the problem, and external validity, they might not be representative

stable-baseline design: behaviour is observed for an extended baseline period before treatment

multiple-baseline design: researchers stagger their introduction of an

intervention ~~over~~ across a variety of contexts, times, or situations

reversal design: behaviour is observed with and without treatment, treatment is taken away for a while (reversal period)

a single-N design studies one participant